In [4]:
```
import pandas as pd
import numpy as np
import requests
```

In [5]:
```
df=pd.read_csv("D:\DSBDA\Assign_3_Mall_Customers.csv")
```

In [6]:
```
df
```

Out[6]:

	CustomerID	Genre	Age Group	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	Teen	19	15	39
1	2	Male	Middle Age	21	15	81
2	3	Female	Middle Age	20	16	6
3	4	Female	Middle Age	23	16	77
4	5	Female	Middle Age	31	17	40
...	...	...	...	...	...	...
195	196	Female	Middle Age	35	120	79
196	197	Female	Elder	45	126	28
197	198	Male	Middle Age	32	126	74
198	199	Male	Middle Age	32	137	18
199	200	Male	Middle Age	30	137	83

200 rows × 6 columns

In [7]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 6 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   CustomerID              200 non-null    int64
 1   Genre                   200 non-null    object
 2   Age Group               200 non-null    object
 3   Age                     200 non-null    int64
 4   Annual Income (k$)      200 non-null    int64
 5   Spending Score (1-100)  200 non-null    int64
dtypes: int64(4), object(2)
memory usage: 9.5+ KB
```

In [8]:
```
df.shape
```

Out[8]:
```
(200, 6)
```

In [9]:
```
df.head
```

Out[9]:
```
<bound method NDFrame.head of      CustomerID   Genre    Age Group  Age  Annual Income (k$)  \
0             1    Male         Teen   19                  15
1             2    Male  Middle Age   21                  15
2             3  Female  Middle Age   20                  16
3             4  Female  Middle Age   23                  16
4             5  Female  Middle Age   31                  17
..          ...     ...         ...  ...                 ...
195         196  Female  Middle Age   35                 120
196         197  Female        Elder   45                 126
197         198    Male  Middle Age   32                 126
198         199    Male  Middle Age   32                 137
199         200    Male  Middle Age   30                 137

     Spending Score (1-100)
0                        39
1                        81
2                         6
3                        77
4                        40
..                      ...
195                      79
196                      28
197                      74
198                      18
199                      83

[200 rows x 6 columns]>
```

In [10]:
```
df.tail
```

Out[10]:
```
<bound method NDFrame.tail of      CustomerID   Genre   Age Group
Age  Annual Income (k$)  \
0             1    Male        Teen   19                  15
1             2    Male  Middle Age   21                  15
2             3  Female  Middle Age   20                  16
3             4  Female  Middle Age   23                  16
4             5  Female  Middle Age   31                  17
..          ...     ...         ...  ...                 ...
195         196  Female  Middle Age   35                 120
196         197  Female       Elder   45                 126
197         198    Male  Middle Age   32                 126
198         199    Male  Middle Age   32                 137
199         200    Male  Middle Age   30                 137

     Spending Score (1-100)
0                        39
1                        81
2                         6
3                        77
4                        40
..                      ...
195                      79
196                      28
197                      74
198                      18
199                      83

[200 rows x 6 columns]>
```

In [11]:
```
df.describe()
```

Out[11]:

	CustomerID	Age	Annual Income (k$)	Spending Score (1-100)
count	200.000000	200.000000	200.000000	200.000000
mean	100.500000	38.850000	60.560000	50.200000
std	57.879185	13.969007	26.264721	25.823522
min	1.000000	18.000000	15.000000	1.000000
25%	50.750000	28.750000	41.500000	34.750000
50%	100.500000	36.000000	61.500000	50.000000
75%	150.250000	49.000000	78.000000	73.000000
max	200.000000	70.000000	137.000000	99.000000

In [12]:
```
df.Age.mean()
```

Out[12]:
```
38.85
```

In [13]:
```
df.Age.mode()
```

Out[13]:
```
0    32
Name: Age, dtype: int64
```

In [14]:
```
df.Age.median()
```

Out[14]:
```
36.0
```

In [15]:
```
df.groupby(['Age']).count()
```

Out[15]:

Age	CustomerID	Genre	Age Group	Annual Income (k$)	Spending Score (1-100)
18	4	4	4	4	4
19	8	8	8	8	8
20	5	5	5	5	5
21	5	5	5	5	5
22	3	3	3	3	3
23	6	6	6	6	6
24	4	4	4	4	4
25	3	3	3	3	3
26	2	2	2	2	2
27	6	6	6	6	6
28	4	4	4	4	4
29	5	5	5	5	5
30	7	7	7	7	7
31	8	8	8	8	8
32	11	11	11	11	11
33	3	3	3	3	3
34	5	5	5	5	5
35	9	9	9	9	9
36	6	6	6	6	6
37	3	3	3	3	3
38	6	6	6	6	6
39	3	3	3	3	3
40	6	6	6	6	6
41	2	2	2	2	2
42	2	2	2	2	2
43	3	3	3	3	3
44	2	2	2	2	2
45	3	3	3	3	3
46	3	3	3	3	3
47	6	6	6	6	6
48	5	5	5	5	5
49	7	7	7	7	7
50	5	5	5	5	5
51	2	2	2	2	2

	CustomerID	Genre	Age Group	Annual Income (k$)	Spending Score (1-100)
Age					
52	2	2	2	2	2
53	2	2	2	2	2
54	4	4	4	4	4
55	1	1	1	1	1
56	1	1	1	1	1
57	2	2	2	2	2
58	2	2	2	2	2
59	4	4	4	4	4
60	3	3	3	3	3
63	2	2	2	2	2
64	1	1	1	1	1
65	2	2	2	2	2
66	2	2	2	2	2
67	4	4	4	4	4
68	3	3	3	3	3
69	1	1	1	1	1
70	2	2	2	2	2

In [16]:
```
df.groupby(['Genre']).count()
```

Out[16]:

	CustomerID	Age Group	Age	Annual Income (k$)	Spending Score (1-100)
Genre					
Female	112	112	112	112	112
Male	88	88	88	88	88

In [17]:
```
df.Age.std()
```

Out[17]:
```
13.969007331558883
```

In [18]:
```
df[['Age' , 'Annual Income (k$)', 'Spending Score (1-100)']].mean()
```

Out[18]:
```
Age                       38.85
Annual Income (k$)        60.56
Spending Score (1-100)    50.20
dtype: float64
```

In [19]:
```
df[['Age' , 'Annual Income (k$)', 'Spending Score (1-100)']].mode()
```

Out[19]:

	Age	Annual Income (k$)	Spending Score (1-100)
0	32.0	54	42.0
1	NaN	78	NaN

In [20]:
```
df[['Age' , 'Annual Income (k$)', 'Spending Score (1-100)']].median(
```

Out[20]:
```
Age                       36.0
Annual Income (k$)        61.5
Spending Score (1-100)    50.0
dtype: float64
```

In [21]:
```
df[['Age' , 'Annual Income (k$)', 'Spending Score (1-100)']].max()
```

Out[21]:
```
Age                        70
Annual Income (k$)        137
Spending Score (1-100)     99
dtype: int64
```

In [22]:
```
df[['Age' , 'Annual Income (k$)', 'Spending Score (1-100)']].std()
```

Out[22]:
```
Age                       13.969007
Annual Income (k$)        26.264721
Spending Score (1-100)    25.823522
dtype: float64
```

In [23]:
```
df2 = df.groupby('Genre')
```

In [24]:
```
df
```

Out[24]:

	CustomerID	Genre	Age Group	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	Teen	19	15	39
1	2	Male	Middle Age	21	15	81
2	3	Female	Middle Age	20	16	6
3	4	Female	Middle Age	23	16	77
4	5	Female	Middle Age	31	17	40
...	...	...	...	...	...	...
195	196	Female	Middle Age	35	120	79
196	197	Female	Elder	45	126	28
197	198	Male	Middle Age	32	126	74
198	199	Male	Middle Age	32	137	18
199	200	Male	Middle Age	30	137	83

200 rows × 6 columns

In [25]:
```
for Genre, Genre_f in df2:
    print(Genre)
    print(Genre_f)
```

```
Female
     CustomerID   Genre   Age Group  Age  Annual Income (k$)  \
2             3  Female  Middle Age   20                  16   
3             4  Female  Middle Age   23                  16   
4             5  Female  Middle Age   31                  17   
5             6  Female  Middle Age   22                  17   
6             7  Female  Middle Age   35                  18   
..          ...     ...         ...  ...                 ...   
191         192  Female  Middle Age   32                 103   
193         194  Female  Middle Age   38                 113   
194         195  Female       Elder   47                 120   
195         196  Female  Middle Age   35                 120   
196         197  Female       Elder   45                 126   

     Spending Score (1-100)  
2                         6  
3                        77  
4                        40  
5                        76  
6                         6  
..                      ...  
191                      69  
193                      91  
194                      16  
195                      79  
196                      28  

[112 rows x 6 columns]
Male
     CustomerID Genre   Age Group  Age  Annual Income (k$)  \
0             1  Male        Teen   19                  15   
1             2  Male  Middle Age   21                  15   
8             9  Male       Elder   64                  19   
10           11  Male       Elder   67                  19   
14           15  Male  Middle Age   37                  20   
..          ...   ...         ...  ...                 ...   
187         188  Male  Middle Age   28                 101   
192         193  Male  Middle Age   33                 113   
197         198  Male  Middle Age   32                 126   
198         199  Male  Middle Age   32                 137   
199         200  Male  Middle Age   30                 137   

     Spending Score (1-100)  
0                        39  
1                        81  
8                         3  
10                       14  
14                       13  
..                      ...  
187                      68  
192                       8  
197                      74  
198                      18  
199                      83  
```

```
[88 rows x 6 columns]
```

In [26]:
```
df2.get_group('Male')
```

Out[26]:

	CustomerID	Genre	Age Group	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	Teen	19	15	39
1	2	Male	Middle Age	21	15	81
8	9	Male	Elder	64	19	3
10	11	Male	Elder	67	19	14
14	15	Male	Middle Age	37	20	13
...	...	...	...	...	...	...
187	188	Male	Middle Age	28	101	68
192	193	Male	Middle Age	33	113	8
197	198	Male	Middle Age	32	126	74
198	199	Male	Middle Age	32	137	18
199	200	Male	Middle Age	30	137	83

88 rows × 6 columns

In [27]:
```
df2.get_group('Female')
```

Out[27]:

	CustomerID	Genre	Age Group	Age	Annual Income (k$)	Spending Score (1-100)
2	3	Female	Middle Age	20	16	6
3	4	Female	Middle Age	23	16	77
4	5	Female	Middle Age	31	17	40
5	6	Female	Middle Age	22	17	76
6	7	Female	Middle Age	35	18	6
...	...	...	...	...	...	...
191	192	Female	Middle Age	32	103	69
193	194	Female	Middle Age	38	113	91
194	195	Female	Elder	47	120	16
195	196	Female	Middle Age	35	120	79
196	197	Female	Elder	45	126	28

112 rows × 6 columns

In [28]:
```
df2[['Age' , 'Annual Income (k$)', 'Spending Score (1-100)']].median
```

Out[28]:

	Age	Annual Income (k$)	Spending Score (1-100)
Genre			
Female	35.0	60.0	50.0
Male	37.0	62.5	50.0

In [29]:
```
df2[['Age' , 'Annual Income (k$)', 'Spending Score (1-100)']].mean()
```

Out[29]:

Genre	Age	Annual Income (k$)	Spending Score (1-100)
Female	38.098214	59.250000	51.526786
Male	39.806818	62.227273	48.511364

In [30]:
```
df2[['Age' , 'Annual Income (k$)', 'Spending Score (1-100)']].max()
```

Out[30]:

Genre	Age	Annual Income (k$)	Spending Score (1-100)
Female	68	126	99
Male	70	137	97

In [31]:
```
df2[['Age' , 'Annual Income (k$)', 'Spending Score (1-100)']].min()
```

Out[31]:

Genre	Age	Annual Income (k$)	Spending Score (1-100)
Female	18	16	5
Male	18	15	1

In [32]:
```
df2[['Age' , 'Annual Income (k$)', 'Spending Score (1-100)']].std()
```

Out[32]:

Genre	Age	Annual Income (k$)	Spending Score (1-100)
Female	12.644095	26.011952	24.11495
Male	15.514812	26.638373	27.89677

In [33]:
```
url = "https://archive.ics.uci.edu/ml/machine-learning-databases/iri
r = requests.get(url)
```

In [34]:
```
df3 = pd.read_csv(url)
```

In [35]:
```
df3
```

Out[35]:

	5.1	3.5	1.4	0.2	Iris-setosa
0	4.9	3.0	1.4	0.2	Iris-setosa
1	4.7	3.2	1.3	0.2	Iris-setosa
2	4.6	3.1	1.5	0.2	Iris-setosa
3	5.0	3.6	1.4	0.2	Iris-setosa
4	5.4	3.9	1.7	0.4	Iris-setosa
...	...	...	...	...	...
144	6.7	3.0	5.2	2.3	Iris-virginica
145	6.3	2.5	5.0	1.9	Iris-virginica
146	6.5	3.0	5.2	2.0	Iris-virginica
147	6.2	3.4	5.4	2.3	Iris-virginica
148	5.9	3.0	5.1	1.8	Iris-virginica

149 rows × 5 columns

In [36]:
```
df3.columns=("A" , "B" , " C " , "D" , "E")
```

In [37]:
```
df3
```

Out[37]:

	A	B	C	D	E
0	4.9	3.0	1.4	0.2	Iris-setosa
1	4.7	3.2	1.3	0.2	Iris-setosa
2	4.6	3.1	1.5	0.2	Iris-setosa
3	5.0	3.6	1.4	0.2	Iris-setosa
4	5.4	3.9	1.7	0.4	Iris-setosa
...	...	...	...	...	...
144	6.7	3.0	5.2	2.3	Iris-virginica
145	6.3	2.5	5.0	1.9	Iris-virginica
146	6.5	3.0	5.2	2.0	Iris-virginica
147	6.2	3.4	5.4	2.3	Iris-virginica
148	5.9	3.0	5.1	1.8	Iris-virginica

149 rows × 5 columns

In [38]:
```
df4 =df3.groupby("E")
```

In [39]:
```
df4
```

Out[39]:
```
<pandas.core.groupby.generic.DataFrameGroupBy object at 0x000001EF3
51D8D30>
```

In [41]:
```
df4.get_group("Iris-setosa")
```

Out[41]:

	A	B	C	D	E
0	4.9	3.0	1.4	0.2	Iris-setosa
1	4.7	3.2	1.3	0.2	Iris-setosa
2	4.6	3.1	1.5	0.2	Iris-setosa
3	5.0	3.6	1.4	0.2	Iris-setosa
4	5.4	3.9	1.7	0.4	Iris-setosa
5	4.6	3.4	1.4	0.3	Iris-setosa
6	5.0	3.4	1.5	0.2	Iris-setosa
7	4.4	2.9	1.4	0.2	Iris-setosa
8	4.9	3.1	1.5	0.1	Iris-setosa
9	5.4	3.7	1.5	0.2	Iris-setosa
10	4.8	3.4	1.6	0.2	Iris-setosa
11	4.8	3.0	1.4	0.1	Iris-setosa
12	4.3	3.0	1.1	0.1	Iris-setosa
13	5.8	4.0	1.2	0.2	Iris-setosa
14	5.7	4.4	1.5	0.4	Iris-setosa
15	5.4	3.9	1.3	0.4	Iris-setosa
16	5.1	3.5	1.4	0.3	Iris-setosa
17	5.7	3.8	1.7	0.3	Iris-setosa
18	5.1	3.8	1.5	0.3	Iris-setosa
19	5.4	3.4	1.7	0.2	Iris-setosa
20	5.1	3.7	1.5	0.4	Iris-setosa
21	4.6	3.6	1.0	0.2	Iris-setosa
22	5.1	3.3	1.7	0.5	Iris-setosa
23	4.8	3.4	1.9	0.2	Iris-setosa
24	5.0	3.0	1.6	0.2	Iris-setosa
25	5.0	3.4	1.6	0.4	Iris-setosa
26	5.2	3.5	1.5	0.2	Iris-setosa
27	5.2	3.4	1.4	0.2	Iris-setosa
28	4.7	3.2	1.6	0.2	Iris-setosa
29	4.8	3.1	1.6	0.2	Iris-setosa
30	5.4	3.4	1.5	0.4	Iris-setosa
31	5.2	4.1	1.5	0.1	Iris-setosa
32	5.5	4.2	1.4	0.2	Iris-setosa
33	4.9	3.1	1.5	0.1	Iris-setosa
34	5.0	3.2	1.2	0.2	Iris-setosa

	A	B	C	D	E
35	5.5	3.5	1.3	0.2	Iris-setosa
36	4.9	3.1	1.5	0.1	Iris-setosa
37	4.4	3.0	1.3	0.2	Iris-setosa
38	5.1	3.4	1.5	0.2	Iris-setosa
39	5.0	3.5	1.3	0.3	Iris-setosa
40	4.5	2.3	1.3	0.3	Iris-setosa
41	4.4	3.2	1.3	0.2	Iris-setosa
42	5.0	3.5	1.6	0.6	Iris-setosa
43	5.1	3.8	1.9	0.4	Iris-setosa
44	4.8	3.0	1.4	0.3	Iris-setosa
45	5.1	3.8	1.6	0.2	Iris-setosa
46	4.6	3.2	1.4	0.2	Iris-setosa
47	5.3	3.7	1.5	0.2	Iris-setosa
48	5.0	3.3	1.4	0.2	Iris-setosa

In [42]:
```
df4.get_group("Iris-virginica")
```

Out[42]:

	A	B	C	D	E
99	6.3	3.3	6.0	2.5	Iris-virginica
100	5.8	2.7	5.1	1.9	Iris-virginica
101	7.1	3.0	5.9	2.1	Iris-virginica
102	6.3	2.9	5.6	1.8	Iris-virginica
103	6.5	3.0	5.8	2.2	Iris-virginica
104	7.6	3.0	6.6	2.1	Iris-virginica
105	4.9	2.5	4.5	1.7	Iris-virginica
106	7.3	2.9	6.3	1.8	Iris-virginica
107	6.7	2.5	5.8	1.8	Iris-virginica
108	7.2	3.6	6.1	2.5	Iris-virginica
109	6.5	3.2	5.1	2.0	Iris-virginica
110	6.4	2.7	5.3	1.9	Iris-virginica
111	6.8	3.0	5.5	2.1	Iris-virginica
112	5.7	2.5	5.0	2.0	Iris-virginica
113	5.8	2.8	5.1	2.4	Iris-virginica
114	6.4	3.2	5.3	2.3	Iris-virginica
115	6.5	3.0	5.5	1.8	Iris-virginica
116	7.7	3.8	6.7	2.2	Iris-virginica
117	7.7	2.6	6.9	2.3	Iris-virginica
118	6.0	2.2	5.0	1.5	Iris-virginica
119	6.9	3.2	5.7	2.3	Iris-virginica
120	5.6	2.8	4.9	2.0	Iris-virginica
121	7.7	2.8	6.7	2.0	Iris-virginica
122	6.3	2.7	4.9	1.8	Iris-virginica
123	6.7	3.3	5.7	2.1	Iris-virginica
124	7.2	3.2	6.0	1.8	Iris-virginica
125	6.2	2.8	4.8	1.8	Iris-virginica
126	6.1	3.0	4.9	1.8	Iris-virginica
127	6.4	2.8	5.6	2.1	Iris-virginica
128	7.2	3.0	5.8	1.6	Iris-virginica
129	7.4	2.8	6.1	1.9	Iris-virginica
130	7.9	3.8	6.4	2.0	Iris-virginica
131	6.4	2.8	5.6	2.2	Iris-virginica
132	6.3	2.8	5.1	1.5	Iris-virginica
133	6.1	2.6	5.6	1.4	Iris-virginica

	A	B	C	D	E
134	7.7	3.0	6.1	2.3	Iris-virginica
135	6.3	3.4	5.6	2.4	Iris-virginica
136	6.4	3.1	5.5	1.8	Iris-virginica
137	6.0	3.0	4.8	1.8	Iris-virginica
138	6.9	3.1	5.4	2.1	Iris-virginica
139	6.7	3.1	5.6	2.4	Iris-virginica
140	6.9	3.1	5.1	2.3	Iris-virginica
141	5.8	2.7	5.1	1.9	Iris-virginica
142	6.8	3.2	5.9	2.3	Iris-virginica
143	6.7	3.3	5.7	2.5	Iris-virginica
144	6.7	3.0	5.2	2.3	Iris-virginica
145	6.3	2.5	5.0	1.9	Iris-virginica
146	6.5	3.0	5.2	2.0	Iris-virginica
147	6.2	3.4	5.4	2.3	Iris-virginica
148	5.9	3.0	5.1	1.8	Iris-virginica

In [43]:
```
df4.mean()
```

Out[43]:

	A	B	C	D
E				
Iris-setosa	5.004082	3.416327	1.465306	0.244898
Iris-versicolor	5.936000	2.770000	4.260000	1.326000
Iris-virginica	6.588000	2.974000	5.552000	2.026000

In [44]:
```
df4.std()
```

Out[44]:

	A	B	C	D
E				
Iris-setosa	0.355879	0.384787	0.175061	0.108130
Iris-versicolor	0.516171	0.313798	0.469911	0.197753
Iris-virginica	0.635880	0.322497	0.551895	0.274650

In [ ]: